

ORANGE-CARITAT AB, France

WMO: 075790

Lat: 44.144N

Lon: 4.861E

Elev: 197

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.3	28.9	15.3	12.1	32.5	19.0	14.4	35.1	39.9	42.1	36.9	40.3	8.7	350	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.0	94.3	69.9	91.4	69.3	88.8	68.6	72.8	87.6	71.5	85.9	70.2	84.2	8.7	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	105.6	77.5	67.0	100.0	76.8	65.5	94.9	76.0	36.5	87.4	35.3	86.3	34.2	84.4	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.1	30.5	27.3	21.8	99.0	3.7	3.5	19.2	101.5	17.0	103.5	14.9	105.5	12.2	108.0		
(5)			20.6	75.7	3.9	1.4	17.8	76.7	15.5	77.5	13.3	78.3	10.4	79.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	42.8	44.7	51.1	56.2	63.6	71.4	76.0	75.0	66.9	59.7	49.7	43.8		
	DBStd	12.97	5.92	6.08	5.58	5.55	5.44	5.88	5.00	5.08	5.35	5.75	6.29	6.14		
	HDD50	752	231	163	54	11	0	0	0	0	0	5	80	208		
	HDD65	3395	687	569	432	268	89	12	2	2	40	178	460	656		
	CDD50	3850	10	14	87	197	422	641	804	774	508	307	70	16		
	CDD65	1018	0	0	0	4	46	203	342	312	97	14	0	0		
	CDH74	10149	0	0	3	65	473	2070	3579	3062	823	74	0	0		
	CDH80	4159	0	0	0	6	100	819	1638	1364	227	5	0	0		
(14) Wind	WSAvg	10.4	11.1	12.1	11.3	10.5	10.0	10.1	10.1	9.6	9.6	9.1	10.8	10.3		
(15) Precipitation	PrecAvg	29.3	2.3	1.6	1.6	2.6	2.3	1.7	1.5	1.7	4.4	3.9	3.9	1.9		
	PrecMax	47.4	6.5	4.8	4.0	6.6	5.2	4.7	5.1	5.2	13.5	6.9	10.7	8.3		
	PrecMin	19.2	0.0	0.0	0.0	0.5	0.2	0.1	0.0	0.1	0.8	0.9	0.9	0.0		
	PrecStd	6.9	1.7	1.3	1.2	1.5	1.4	1.2	1.3	1.1	2.9	1.7	2.6	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.7	65.3	73.5	79.9	87.5	95.9	97.6	99.2	90.5	79.4	68.2	62.1		
		MCWB	53.2	52.7	55.6	61.6	66.1	71.2	70.6	71.2	68.2	64.7	59.7	54.7		
	2%	DB	57.7	61.2	68.6	75.4	82.6	91.6	94.1	94.4	85.8	75.9	65.3	58.9		
		MCWB	50.9	50.9	54.7	59.3	64.2	69.6	69.8	69.8	66.6	64.3	57.9	52.9		
	5%	DB	55.1	58.1	65.5	71.7	79.1	88.2	91.5	90.9	82.2	73.1	62.8	56.4		
		MCWB	49.2	49.4	53.2	57.5	62.6	68.3	69.3	69.4	65.3	63.0	56.3	51.2		
	10%	DB	52.5	55.1	62.3	68.2	75.9	84.7	88.9	87.8	78.9	70.3	60.2	53.8		
		MCWB	47.3	47.9	51.7	55.5	61.4	66.9	68.5	68.9	64.5	61.4	54.8	49.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.2	54.0	57.9	63.2	69.0	74.6	74.1	74.4	71.1	67.6	61.7	56.0		
		MCDB	59.7	61.9	68.2	76.4	83.0	92.7	89.3	87.9	81.6	74.9	66.2	60.0		
	2%	WB	51.9	52.1	55.8	60.6	65.9	71.8	72.6	72.8	69.3	65.9	59.0	53.8		
		MCDB	56.9	59.1	65.8	73.2	78.9	86.7	88.0	86.8	79.7	73.5	63.8	58.0		
	5%	WB	50.0	50.6	54.3	58.5	64.2	69.9	71.2	71.5	67.7	64.2	57.3	52.0		
		MCDB	54.0	56.7	63.3	69.0	75.8	84.4	86.4	85.5	77.9	71.0	61.8	55.5		
	10%	WB	48.1	48.5	52.7	56.7	62.5	68.2	69.8	70.1	66.0	62.4	55.4	49.8		
		MCDB	51.8	54.1	60.7	65.8	73.2	82.1	85.0	83.5	75.9	68.5	59.5	53.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	16.3	19.2	19.9	20.5	22.2	23.0	22.9	20.6	17.6	14.7	13.7		
		MCDBR	17.9	22.3	24.6	26.0	26.3	27.7	27.6	28.9	26.9	21.6	18.2	16.4		
		MCWBR	12.1	13.6	13.3	12.9	11.7	11.0	9.7	10.1	11.3	11.6	11.9	11.2		
	5% WB	MCDBR	16.3	20.3	22.4	23.6	24.2	26.3	25.9	26.2	23.5	19.5	16.1	14.6		
		MCWBR	11.6	13.1	12.8	12.4	11.8	11.1	10.2	10.4	11.2	11.1	11.2	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.342	0.381	0.398	0.407	0.406	0.399	0.397	0.385	0.387	0.348	0.318		
	taud		2.454	2.392	2.297	2.276	2.300	2.329	2.349	2.362	2.383	2.368	2.430	2.472		
	Ebn at Noon		254	268	270	274	274	275	272	272	266	247	240	243		
	Edh at Noon		23	29	36	40	41	39	38	36	33	30	24	21		
(44) All-Sky Solar Radiation	RadAvg		524	824	1229	1594	1913	2234	2235	1919	1462	916	574	454		
	RadStd		57	83	111	138	148	109	116	76	92	90	73	51		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	+1.69	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page